

Boba â€” Feature Status (all components)

Revision: 7 **Last modified:** 2026-06-16T23:30:00Z **Scope:** Every system component, service, infrastructure piece, and client app of the Boba project â€” one row per REAL unit (endpoint / handler / client method / component control / plugin / subcommand / script), grouped by component. **Authority:** assembled by READ-ONLY repo inventory (codegraph + grep + source reading) on 2026-06-15, expanded to per-unit granularity (Â§11.4.118 discovery-pressure) on 2026-06-16. Cross-references AGENTS.md, CLAUDE.md, docs/REMAINING_WORK_PLAN.md.

Rev 7 (2026-06-16) â€” live-PASS update for the search/auth fix batch (HEAD 7e9cab5): the multi-word URL-encoding fix (da7d709, ~17 nova3 plugins), the new RUTRACKER_COOKIES injection (2fc29fc), NNMCLUB_COOKIES auth, /auth/status cookie reflection (9c2f8dc), the new GET /api/v1/healthz JSON endpoint (137d7ff), and Â§11.4.85 stress+chaos coverage for the multi-word fix (7e9cab5) were all PROVEN end-to-end on the live nezha stack â€” full-fleet the matrix = **2600 results / 23 contributing trackers / zero** plugin_bad_query_encoding, all four private trackers authenticate (rutracker 50, nnmclub 50, kinozal 50, iptorrents 49). Evidence: docs/qa/search-fix-verify-20260616/.kickass.py reclassified **Wonâ€™t-fix structurally-impossible (Â§11.4.112)** per docs/research/kickass_403_20260616/. NOTE (Â§11.4.6): no Go files were touched this session â€” the Go rows are unchanged (the pre-existing magnet_stress_chaos_test.go guard stands; no NEW Go query-encoding guard was added this session, so none is claimed here).

Captured-evidence-driven (Â§11.4.5 / Â§11.4.45 / Â§11.4.86). Every feature row cites a real source file (file:line where load-bearing), endpoint, command, or control. **No invented features (Â§11.4.6).** As of Rev 5 the **Video** column is DEFINITIVE for every row â€” there is no bare PENDING. Each cell is exactly one of: VIDEO-CONFIRMED â€” <file> (the feature is shown on-screen in a committed recording), PENDING (UI â€” film next) (a user-visible control/dialog not yet individually filmed), or N/A (no UI â€” test-covered + exercised by <journey>) (a non-user-visible unit â€” REST endpoint / handler / client method / parser / scanner / service / crypto / repo / script â€” which has no screen of its own and is confirmed by its cited tests plus the UI/CLI journey that drives it). **Honest classification (Â§11.4.6/Â§11.4.143): a row is VIDEO-CONFIRMED only when a real recording actually shows it.**

Per-row Video tally (288 feature rows): **28 VIDEO-CONFIRMED, 14 PENDING (UI â€” film next), 246 N/A (no UI â€” test-covered).**

Column legend: - **Impl** = implemented / partial / stub / missing - **Wired** = reachable on a running default-profile stack (yes / no / opt-in / host) - **Tests** = test types present + cite file (unit / integration / e2e / security / stress-chaos / property / contract); none if no covering test found - **Validation** = honest validation posture (tested-green-in-suite / unit-only / live-evidence-captured / not-validated / operator-blocked) - **Video** = DEFINITIVE video-recording status â€” one of VIDEO-CONFIRMED â€” <file> / PENDING (UI â€” film next) / N/A (no UI â€” test-covered + exercised by <journey>)

Video-recording confirmations (Â§11.4.143 / Â§11.4.83)

Real-use recordings (Claude-vision analyzed â€” the HelixAgent ensemble vision is a proven stub, see docs/qa/recording-readiness-20260615/) in /Volumes/T7/ Downloads/Recordings/ (project-prefix boba-). Per-row Video cells stay PENDING where a feature has not been *individually* filmed; the recordings below confirm the headline user-facing flows on-screen.

Recording	Surface	Features confirmed	Verdict
boba-cli-orchestrator-demo.mp4	CLI (boba-ctl)	status / health / list â€” 4 services running & healthy	docs/qa/recordings-20260615/boba-cli-verdict.md
boba-web-search-flow.mp4	Web dashboard	launch â†’ search â€œdebianâ€” â†’ live results â†’ â€œ829 results (288 merged)â€” complete dashboard load (29 trackers, qBit connected), result rows render qBit + Download buttons (clickable), tab nav (Results/Active Downloads/Trackers/Schedules/Hooks), Jackett / jackett credentials page, theme content-script detects a magnet + appends a ðŸŒŽ MAGNET badge (non-torrent correctly skipped); popup UI	boba-web-search-flow-verdict.md
boba-web-feature-tour.mp4 + boba-web-dashboard-tour.mp4	Web dashboard		boba-web-feature-tour-verdict.md
boba-extension-{scan-detect,popup,options,tour}.mp4	BobaLink extension (MV3, real --load-extension Chromium)		boba-extension-verdict.md

Recording	Surface	Features confirmed	Verdict
boba-web-tabs-theme-tour.mp4 (+ frames boba-web-tab-active-downloads.png / boba-web-tab-schedules.png / boba-web-tab-hooks.png / boba-web-theme-toggle.png) + boba-web-jackett-add-credential.png	Web dashboard + Jackett page	(header, server status, Refresh + Send All, Detected-torrents); options page (7 settings tabs + server config) Active Downloads / Schedules / Hooks tabs each render their empty-state + columns; theme toggle; processing spinner (â€œFound N resultsâ€œ); bridge/qBit-Connected header indicator; Jackett Add-credential dialog (Name/Username/Password/Cookies + Cancel/Save)	Claude-vision verified 2026-06-16

Honest scope (Â§11.4.6, no bluff): - **Backend services** (download-proxy / merge / boba-jackett / Go) verified **200 server-side inside the VM** and exercised through the web UI recordings. - **Console errors** seen in the web recordings are an **SSH-tunnel SSE/poll artifact** (macOSâ€™ VM), NOT product defects â€” every endpoint is 200 in the VM. See the feature-tour verdict. - **BobaLink extension:** â€” **VIDEO-CONFIRMED** â€” the prior -- load-extension tooling gap was closed by building extension/scripts/record-features.mjs (loads the built MV3 artifact into a real Chromium, records the scan/popup/options journeys). Plus its passing test suite. - **TUI / mobile / desktop:** do not exist in this project (server-side + Angular web + browser extension only) â€” nothing to record.

Component summary

#	Component	Service / Port	Features cataloged
1	Download Proxy + Merge Search	qbittorrent-proxy :7186 / :7187	69

#	Component Service (Python/ FastAPI)	Service / Port	Features cataloged
2	qBitTorrent-go backend (Go/Gin)	qbittorrent-proxy-go :7186/:7187/:7188 (opt-in -- profile go)	47
3	boba-jackett (Go)	boba-jackett :7189	26
4	Tracker plugins (plugins/*.py)	run inside qbittorrent- proxy	30
5	Angular 21 frontend dashboard (frontend/)	served from :7187	34
6	BobaLink browser extension (extension/)	WXT MV3 client	39
7	WebUI bridge (webui- bridge.py)	host process :7188	4
8	Infrastructure / CLI / shell scripts	host	40

Total features cataloged: 289 (Rev 7: +1 â€” new GET /api/v1/healthz endpoint)

1. Download Proxy + Merge Search Service (Python/ FastAPI) â€” : 7186 / : 7187

Entry: download-proxy/src/main.py (starts legacy proxy thread + uvicorn api : app).
This is the **shipped default product**.

1a. Merge Search Service â€” REST API (download-proxy/src/api/ routes.py)

Feature	Impl	Wired Tests	Validation	Video
POST /api/v1/ search â€” async fan-out search, returns search_id (routes.py:283)	implemented yes	integration tests/integration/ test_merge_api.py; e2e test_full_pipeline.py; stress test_merge_search_stress_chaos.py	tested- green-in- suite + LIVE	VID 2020 jour foun web verd rec bob flo
POST /api/v1/ search/sync â€” synchronous search (routes.py:377)	implemented yes	integration test_merge_api.py; stress test_search_stress.py	tested- green-in- suite; resets over tunnel (RW-08)	N/A + ex flow
	implemented yes			

Feature	Impl	Wired Tests	Validation	Video
GET /api/v1/ search/stream/ {id} â€” SSE result stream (routes.py:547)		integration test_realtime_streaming.py, test_streaming_browser.py	tested- green-in- suite	N/A + ex flow
GET /api/v1/ search/{id} â€” poll search state (routes.py:586)	implemented yes	integration test_merge_api.py	tested- green-in- suite	N/A + ex flow
POST /api/v1/ search/{id}/ abort â€” cancel in- flight search (routes.py:619)	implemented yes	integration test_merge_api.py	tested- green-in- suite	N/A + ex flow
GET /api/v1/ downloads/ active â€” qBit active torrents (routes.py:629)	implemented yes	integration test_buttons_api.py	tested- green-in- suite	N/A + ex dash tab)
POST /api/v1/ auth/ qbittorrent â€” qBit WebUI login (routes.py:671)	implemented yes	integration test_login_actions.py	tested- green-in- suite	N/A + ex login
POST /api/v1/ download â€” add torrent to qBit (magnet/url) (routes.py:958)	implemented yes	integration test_button_functions.py, test_buttons_api.py; security test_ssrf_and_magnet_auth.py	tested- green-in- suite; SSRF gap RW-03	N/A + ex row
POST /api/v1/ download/upload â€” upload .torrent file to qBit (routes.py:1123)	implemented yes	unit tests/unit/api_layer/ test_download_upload.py	unit-only + integration	N/A + ex path
POST /api/v1/ download/file â€” fetch tracker .torrent w/ auth cookies (routes.py:1212)	implemented yes	integration test_buttons_api.py; security test_ssrf_and_magnet_auth.py	tested- green-in- suite; SSRF gap RW-03	N/A + ex row
POST /api/v1/ magnet â€” resolve result â†’ magnet link (routes.py:1279)	implemented yes	integration test_magnet_dialog.py, test_buttons_api.py	tested- green-in- suite; auth gap RW-04	N/A + ex dialo
GET /api/v1/ theme â€” read theme state (routes.py:61)	implemented yes implemented yes	contract test_crossapp_theme_contract.py	tested- green-in- suite	N/A + ex pick

Feature	Impl	Wired Tests	Validation	Video
PUT /api/v1/ theme "persist theme (routes.py:67)		e2e test_theme_runtime.py, test_crossapp_theme.py	tested- green-in- suite	N/A + ex pick
GET /api/v1/ theme/stream "SSE theme broadcast (routes.py:76)	implemented yes	e2e test_crossapp_theme.py	tested- green-in- suite	N/A + ex pick

1b. Auth API " per-tracker (download-proxy/src/api/auth.py)

Feature	Impl	Wired Tests	Validation
GET /auth/ rutracker/ status "RuTracker session state (auth.py:56)	implemented yes	integration test_auth_and_links.py; e2e test_live_stack_evidence.py	partial " CAPTCHA operator-blocked (BOB-008)
GET /auth/ rutracker/ captcha "fetch CAPTCHA image (auth.py:108)	implemented yes	integration test_auth_and_links.py	partial " operator- blocked (BOB-008)
POST /auth/ rutracker/ login "username/ password + CAPTCHA login (auth.py:214)	implemented yes	integration test_auth_and_links.py	partial " operator- blocked (BOB-008)
POST /auth/ rutracker/ cookie-login "cookie-based login (auth.py:298)	implemented yes	integration test_auth_and_links.py	PASS " LIVE 2026-06-16 "RUTRACKER_COOKIE cookie path authenticat (2fc29fc); rutracker the matrix = 50 rea results on nezha (docs qa/search-fix- verify-20260616/ Â§1)
GET /auth/ nnmclub/ status "NNM-Club session state (auth.py:350)	implemented yes	e2e test_live_stack_evidence.py:265 (skips " SOURCE"ARTIFACT drift RW-07)	partial " route 404 on running container (RW-07)
	implemented yes	e2e test_live_stack_evidence.py	partial " live-gated

Feature	Impl	Wired Tests		Validation
POST /auth/ nnmclub/ login â€” NNM-Club login (auth.py:407)				PASS â€” LIVE 2026-06-16 â€” now reflects RUTRACKER_COOKIE NNMCLUB_COOKIES env before first search (9c2f8dc), so dashboard chips show rutracker/nnmclub green (docs/qa/search- fix- verify-20260616/ Â§3)
GET /auth/ status â€” all- tracker aggregate session state (auth.py:479)	implemented yes		integration test_auth_state_ui.py; auth-status cookie-reflection test (Â§1.1)	
POST /auth/ qbittorrent/ logout â€” qBit WebUI logout (auth.py:532)	implemented yes		integration test_login_actions.py	tested-green-in-suite
Kinozal auth (status/login) REST route	missing	no	none â€” no kinozal/login route in auth.py	not-implemented (search path uses cookies via orchestrator)
IPTorrents auth (status/login) REST route	missing	no	none â€” no iptorrents/login route in auth.py	not-implemented (env creds + cookie flow via Jackett UI)

1c. Search orchestration internals (download-proxy/src/ merge_service/search.py)

Feature	Impl	Wired Tests
Concurrency caps MAX_CONCURRENT_SEARCHES (429 on saturation)	implemented yes	concurrency test_orchestrator_se
Per-search tracker semaphore MAX_CONCURRENT_TRACKERS	implemented yes	concurrency test_tracker_semaphore

Feature**Impl****Wired Tests**

Public-tracker subprocess fan-out (NDJSON
via patched novaprinter)

implemented yes

e2e
test_public_trackers_return_
stress
test_search_orchestration_st

Public-tracker deadline kill
PUBLIC_TRACKER_DEADLINE_SECONDS

implemented yes

chaos test_tracker_failure.py

Private-tracker aiohttp+cookie path
(rutracker/kinozal/nnmclub/iptorrents)

implemented yes

integration test_tracker_auth_liv

Per-tracker TrackerSearchStat real-
time stats

implemented yes

contract test_tracker_stats_cont
property test_tracker_stats_prop

Stderr error classification
(_classify_plugin_stderr)

implemented yes

chaos test_tracker_failure.py

Feature	Impl	Wired Tests
TTL-bounded data stores (cachetools) â€” no leak	implemented yes	memory test_orchestrator_cache

1d. Deduplication (download-proxy/src/merge_service/deduplicator.py)

Feature	Impl	Wired Tests	Validation	Video
Tier 1 dedup â€” metadata (imdb/tmdb) key	implemented yes	property test_deduplicator_properties.py; benchmark test_deduplication_benchmark.py	tested-green-in-suite	N/A (no UI â€” test-covered + exercised by web search flow â€œ288 mergedâ€” result)
Tier 2 dedup â€” infohash key	implemented yes	property test_deduplicator_properties.py	tested-green-in-suite	N/A (no UI â€” test-covered + exercised by web search flow â€œ288 mergedâ€” result)
Tier 3 dedup â€” name+size key	implemented yes	property test_deduplicator_properties.py	tested-green-in-suite	N/A (no UI â€” test-covered + exercised by web search flow â€œ288 mergedâ€” result)
Tier 4 dedup â€” fuzzy Levenshtein name match	implemented yes	property test_deduplicator_properties.py; benchmark test_deduplication_benchmark.py	tested-green-in-suite	N/A (no UI â€” test-covered + exercised by web search flow â€œ288

Feature	Impl	Wired Tests	Validation Video
			mergedâ€¢  result)

1e. Enrichment “ per metadata source (download-proxy/src/merge_service/enricher.py)

Feature	Impl	Wired Tests	Validation	Vi
Enrichment: OMDb (enricher.py:113)	implemented yes	stress test_enricher_stress_chaos.py	tested-green-in-suite; needs OMDB_API_KEY	N, U te co + ex by se flo re qu co N, U te co + ex by se flo re qu co N, U te co + ex by se flo re qu co N, U te co + ex by se
Enrichment: TMDB	implemented yes	stress test_enricher_stress_chaos.py	tested-green; needs TMDB_API_KEY	N, U te co + ex by se flo re qu co N, U te co + ex by se
Enrichment: TVMaze	implemented yes	stress test_enricher_stress_chaos.py	tested-green-in-suite	N, U te co + ex by se flo re qu co N, U te co + ex by se
Enrichment: AniList	implemented yes	stress test_enricher_stress_chaos.py	tested-green-in-suite	N, U te co + ex by se

Feature	Impl	Wired Tests	Validation	Video
Enrichment: MusicBrainz	implemented yes	stress test_enricher_stress_chaos.py suite	tested-green-in-	fl re qu co N/ U te co + ex by se fl re qu co N/ U te co + ex by se fl re qu co
Enrichment: OpenLibrary	implemented yes	stress test_enricher_stress_chaos.py suite	tested-green-in-	

1f. Tracker validation (download-proxy/src/merge_service/validator.py)

Feature	Impl	Wired	Tests	Validation	Video
Tracker validation HTTP scrape BEP 48 (validator.py:101)	implemented yes		none dedicated found (covered via merge tests)	not-validated (no dedicated test)	N/A (no UI â€” exercised by web search flow seeds/ leechers column) N/A (no UI â€” exercised by web search flow seeds/ leechers column)
Tracker validation UDP scrape BEP 15 (validator.py:162)	implemented yes		none dedicated found	not-validated (no dedicated test)	

1g. Hooks (download-proxy/src/api/hooks.py + merge_service/hooks.py)

Feature	Impl	Wired Tests	Validation
GET /api/v1/hooks â€” list hooks (hooks.py:105)	implemented yes	unit test_hooks_coverage.py; concurrency test_hooks_race.py; security test_hooks_schedules_auth.py	tested- green-in- suite; auth gap RW-01
POST /api/v1/hooks â€” create hook (hooks.py:111)	implemented yes	unit test_hooks_coverage.py; concurrency test_hooks_race.py	tested- green-in- suite; auth gap RW-01
DELETE /api/v1/hooks/{id} â€” delete hook (hooks.py:167)	implemented yes	unit test_hooks_coverage.py	tested- green-in- suite; auth gap RW-01
GET /api/v1/hooks/logs â€” hook execution logs (hooks.py:179)	implemented yes	unit test_hooks_coverage.py	tested- green-in- suite
Hook event dispatch (8 events: search_start..validation_complete) (hooks.py:32)	implemented yes	concurrency test_hooks_race.py	tested- green-in- suite
Hook subprocess execution + sandbox (merge_service/hooks.py)	implemented yes	security test_no_shell_injection.py	tested- green; sandbox hardening RW-01

Feature	Impl	Wired Tests	Validation
1h. Scheduler (download-proxy/src/api/scheduler.py + merge_service/scheduler.py)			

Feature	Impl	Wired Tests	Validation	Video
GET /api/v1/schedules â€” list schedules (scheduler.py:39)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite; auth gap RW-02	N/A (no UI â€” test-covered - exercised by web dashboard (Schedule tab)
POST /api/v1/schedules â€” create schedule (scheduler.py:64)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite; auth gap RW-02	N/A (no UI â€” test-covered - exercised by web dashboard (Schedule tab)
GET /api/v1/schedules/{id} â€” get schedule (scheduler.py:87)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite	N/A (no UI â€” test-covered - exercised by web dashboard (Schedule tab)
PATCH /api/v1/schedules/{id} â€” update schedule (scheduler.py:108)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite	N/A (no UI â€” test-covered - exercised by web dashboard (Schedule tab)
DELETE /api/v1/schedules/{id} â€” delete schedule (scheduler.py:126)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite	N/A (no UI â€” test-covered - exercised by web dashboard

Feature	Impl	Wired Tests	Validation	Video
Scheduler engine (load/save/tick driver) (merge_service/scheduler.py)	implemented yes	integration test_dashboard_automation.py	tested-green-in-suite	Schedule tab) N/A (no UI “ test- covered, internal tick driver)

1i. App-level routes / health / config / dashboard (download-proxy/src/api/__init__.py)

Feature	Impl	Wired Tests	Validation	Video
GET /api/v1/healthz “machine-parsable JSON health probe (routes.py:37)	implemented yes	integration test_merge_api.py	PASS “LIVE 2026-06-16 “NEW endpoint (137d7ff): a bare /healthz was swallowed by the SPA catch-all; now mounted under /api/v1 returning JSON (Â§11.4.43)	N/A (no “ JSON health pr test-cove
GET /health (__init__.py:168)	implemented yes	integration test_live_containers.py	tested-green-in-suite	N/A (no “ test- covered exercised boba-ctl CLI)
GET /api/v1/bridge/health (__init__.py:173)	implemented yes	integration test_bridge_root_liveness.py	tested-green-in-suite	N/A (no “ test- covered exercised web brid health indicator
GET /api/v1/config (__init__.py:247)	implemented yes	integration test_merge_api.py	tested-green-in-suite	N/A (no “ test- covered exercised

Feature	Impl	Wired Tests	Validation	Video
GET /api/v1/stats (__init__.py:276)	implemented yes	observability test_metrics_exist.py	tested- green-in- suite	web dash bootstrap N/A (no â€” test- covered exercised web trac stat dialo
Jinja2 dashboard GET / (__init__.py:337)	implemented yes	integration test_dashboard_rendering.py	tested- green-in- suite	N/A (rou serves th dashboar SPA; the dashboar itself is VIDEO- CONFIR boba-w featur tour.m
Jinja2 dashboard GET /dashboard (__init__.py:342)	implemented yes	integration test_dashboard_rendering.py	tested- green-in- suite	N/A (rou serves th dashboar SPA; the dashboar itself is VIDEO- CONFIR boba-w featur tour.m
SPA catch-all GET / {path:path} (__init__.py:347)	implemented yes	integration test_dashboard_rendering.py	tested- green-in- suite	N/A (rou serves S deep link incl.Â / jacket VIDEO- CONFIR via boba web- featur tour.m

1j. Supporting subsystems (Python)

Feature	Impl	Wired Tests	Va
Retry helpers (merge_service/ retry.py)	implemented yes	covered indirectly	uni

Feature	Impl	Wired Tests	Validation	Video
Jackett autoconfig (Python, merge_service/jackett_autoconfig.py) implemented yes â€” legacy; canonical is boba-jackett		contract test_jackett_autoconfig_contract.py; tes e2e test_jackett_autoconfig_e2e.py; benchmark test_jackett_autoconfig_perf.py		gre sui
Credential scrubbing log filter (config/log_filter.py)	implemented yes	security test_credential_scrubbing.py		tes gre sui

1k. Legacy download proxy (plugins/download_proxy.py :7186)

Feature	Impl	Wired Tests	Validation	Video
HTTP proxy â†’ qBittorrent WebUI w/ tracker URL interception (DownloadHandler :767)	implemented yes	integration test_bridge_header_rewrite.py	tested- green-in- suite	N/A (UI of own â proxy layer; test-cover
Theme injection into proxied WebUI (plugins/theme_injector.py)	implemented yes	integration test_bridge_theme_injection.py	tested- green-in- suite	N/A (UI of own â proxy theme inject test-cover

2. qBitTorrent-go backend (Go/Gin) â€” opt-in --profile go

Entry: qBitTorrent-go/cmd/qbittorrent-proxy/main.go (routes registered :54-101). **Skeleton/rewrite-in-progress** (per AGENTS.md + docs/migration/PARITY_GAPS.md). NOT running by default.

2a. HTTP handlers (internal/api/*.go)

Feature	Impl	Wired Tests	Validation	Video
GET /health â€” HealthHandler (api/health.go:5)	implemented	opt-in go internal/api/api_test.go	unit-only (go test)	N/A UI go-

Feature	Impl	Wired Tests	Validation	Validation
GET /api/v1/config â€” ConfigHandler (api/download.go:210)	implemented	opt-in go coverage_test.go	unit-only	coverage; optional; production; not shipping; defined; N/A
GET /api/v1/bridge/health â€” BridgeHealthHandler (api/download.go:189)	implemented	opt-in go coverage_test.go	unit-only	UI; go-coverage; optional; production; N/A
POST /api/v1/search â€” SearchHandler (api/search.go:14)	implemented (proxies qBit search; no plugin fan-out)	opt-in go api_test.go, coverage_test.go	unit-only	UI; go-coverage; optional; production; N/A
POST /api/v1/search/sync â€” SearchSyncHandler (api/search.go:50)	implemented	opt-in go coverage_test.go	unit-only	UI; go-coverage; optional; production; N/A
GET /api/v1/search/stream/:id â€” SearchStreamHandler SSE (api/search.go:91)	implemented	opt-in go coverage_test.go	unit-only	UI; go-coverage; optional; production; N/A
GET /api/v1/search/:id â€” GetSearchHandler (api/search.go:144)	implemented	opt-in go getsearch_results_test.go, coverage_test.go	unit-only	UI; go-coverage; optional; production; N/A
POST /api/v1/search/:id/abort â€” AbortSearchHandler (api/search.go:177)	implemented	opt-in go coverage_test.go	unit-only	UI; go-coverage; optional; production; N/A
POST /api/v1/download â€” DownloadHandler (api/download.go:17)	partial â€” returns mock	opt-in go coverage_test.go	unit-only; functional hole	N/A; UI; go-

Feature	Impl	Wired Tests	Validation	View
	success, no qBit integration			coverage optional production functional hold N/A UI
POST /api/v1/download/ file â€” DownloadFileHandler (api/download.go:41)	implemented	opt-in go coverage_test.go	unit-only	go- cov opt pro N/A UI
POST /api/v1/magnet â€” MagnetHandler (api/ download.go:86)	implemented	opt-in go api_test.go, magnet_stress_chaos_test.go	unit + stress- chaos (go)	go- cov opt pro N/A UI
GET /api/v1/downloads/ active â€” ActiveDownloadsHandler (api/download.go:147)	stub â€” returns empty array, no qBit query	opt-in go coverage_test.go	unit-only; functional hole	go- cov opt pro stub N/A UI
POST /api/v1/auth/ qbittorrent â€” QBittorrentAuthHandler (api/download.go:156)	implemented	opt-in go coverage_test.go	unit-only	go- cov opt pro N/A UI
GET /api/v1/theme â€” GetThemeHandler (api/ theme.go:90)	implemented	opt-in go theme_hardening_test.go, coverage_test.go	unit-only	go- cov opt pro N/A UI
PUT /api/v1/theme â€” PutThemeHandler (api/ theme.go:96)	implemented	opt-in go theme_hardening_test.go	unit-only	go- cov opt pro N/A UI
GET /api/v1/hooks â€” ListHooksHandler (api/ hooks.go:91)	implemented (store only, no dispatch)	opt-in go api_test.go, coverage_test.go	unit-only	go- cov opt pro N/A UI
POST /api/v1/hooks â€” CreateHookHandler (api/ hooks.go:97)	implemented (store only)	opt-in go api_test.go	unit-only	UI go-

Feature	Impl	Wired Tests	Validation	Video
DELETE /api/v1/hooks/:id â€” DeleteHookHandler (api/hooks.go:109)	implemented (store only)	opt-in go api_test.go	unit-only	coverage: 100% (N/A) UI: none
GET /api/v1/schedules â€” ListSchedulesHandler (api/scheduler_api.go:86)	implemented (store only)	opt-in go coverage_test.go	unit-only; no driver loop (RW-10), schedules never fire	coverage: 100% (N/A) UI: none
POST /api/v1/schedules â€” CreateScheduleHandler (api/scheduler_api.go:92)	implemented (store only)	opt-in go coverage_test.go	unit-only; never fires (RW-10)	coverage: 100% (N/A) UI: none
DELETE /api/v1/schedules/:id â€” DeleteScheduleHandler (api/scheduler_api.go:104)	implemented (store only)	opt-in go coverage_test.go	unit-only	coverage: 100% (N/A) UI: none

2b. qBittorrent Web API client (internal/client/*.go)

Feature	Impl	Wired Tests	Validation	Video
NewClient â€” construct + login (client/client.go:18)	implemented	opt-in go client/client_test.go	unit-only	N/A (no UI â€” go-test-covered client method; opt-in profile)
Login â€” POST /api/v2/auth/login, store SID (client/auth.go:11)	implemented	opt-in go client/auth_test.go	unit-only	N/A (no UI â€” go-test-covered client method;

Feature	Impl	Wired	Tests	Validation	Video
IsAuthenticated â€” SID presence (client/ auth.go:43)	implemented opt-in		go client/ auth_test.go	unit-only	opt-in profile) N/A (no UI â€” go-test- covered client method; opt-in profile) N/A (no UI â€” go-test- covered client method; opt-in profile)
GetSID â€” thread- safe SID accessor (client/ client.go:39)	implemented opt-in		go client/ client_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
GetTorrents â€” GET /api/v2/ torrents/info (client/ torrents.go:13)	implemented opt-in		go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
AddTorrent â€” POST /api/v2/ torrents/add URL (client/ torrents.go:30)	implemented opt-in		go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
AddTorrentFile â€” POST multipart / api/v2/torrents/ add (client/ torrents.go:54)	implemented opt-in		go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
GetAppVersion â€” GET /api/v2/app/ version (client/ torrents.go:87)	implemented opt-in		go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
StartSearch â€” POST /api/v2/	implemented opt-in			unit-only	N/A (no UI â€”

Feature	Impl	Wired	Tests	Validation	Video
search/start (client/ search.go:27)			go client/ coverage_test.go		go-test- covered client method; opt-in profile) N/A (no UI â€”
GetSearchResults â€” GET /api/v2/ search/results (client/ search.go:55)	implemented	opt-in	go client/ coverage_test.go	unit-only	go-test- covered client method; opt-in profile) N/A (no UI â€”
StopSearch â€” POST /api/v2/ search/stop (client/ search.go:84)	implemented	opt-in	go client/ coverage_test.go	unit-only	go-test- covered client method; opt-in profile) N/A (no UI â€”
SearchStatus â€” GET /api/v2/ search/status (client/ search.go:100)	implemented	opt-in	go client/ coverage_test.go	unit-only	go-test- covered client method; opt-in profile) N/A (no UI â€”
ListPlugins â€” GET /api/v2/ search/plugins (client/ search.go:121)	implemented	opt-in	go client/ coverage_test.go	unit-only	go-test- covered client method; opt-in profile)

2c. Services (internal/service/*.go)

Feature	Impl	Wired	Tests
Merge search orchestrator RunSearch poll/dedup loop (service/ merge_search.go:183)	implemented (qBit search only; no python- style plugin fan- out)	opt-in	go service/service_test.go, runsearch_dedup_test.go
StartSearch / GetSearchStatus / AbortSearch lifecycle	implemented	opt-in	go getsearchstatus_hardening_test.go

Feature	Impl	Wired	Tests
(service/ merge_search.go:101-143)			
Dedup key (FileURL+name+size) (service/ merge_search.go:269)	implemented	opt-in	go runsearch_dedup_test.go
FetchTorrent tracker fetch (service/ merge_search.go:276)	stub " returns "not yet implemented"	opt-in	none
Stats counters (service/ merge_search.go:280)	implemented	opt-in	go coverage_test.go
SSE broker pub/sub (service/ sse_broker.go)	implemented (defined but not wired " handlers SSE inline)	opt-in	none dedicated
Metadata enricher (Go)	missing (RW-11)	no	none
Scheduler driver loop (Go)	missing (RW-10) " CRUD only, no cron execution	no	none

2d. webui-bridge binary (Go) + cross-cutting

Feature	Impl	Wired	Tests	Validation	Video
webui-bridge binary (cmd/ webui- bridge/ main.go)	partial (health + root only)	opt-in	none dedicated	not-validated	N/A (no UI of its own " proxy binary; opt- in profile)
CORS + logging middleware (internal/ middleware/)	implemented	opt-in	go middleware/ middleware_test.go	unit-only	N/A (no UI " go-test- covered middleware; opt-in profile)
Models / data types (internal/ models/)	implemented	opt-in	go models/ models_test.go	unit-only	N/A (no UI " go-test- covered data types; opt-in profile)
	implemented	opt-in		unit-only	

Feature	Impl	Wired	Tests	Validation	Video
Env config loader (internal/config/)			go config/ config_test.go		N/A (no UI â€” go-test-covered config loader; opt-in profile)
Log redactor (internal/logging/redactor.go)	implemented	opt-in	go logging/ redactor_test.go	unit-only	N/A (no UI â€” go-test-covered redactor; opt-in profile)

3. boba-jackett (Go) â€” : 7189

Entry: qBittorrent-go/cmd/boba-jackett/main.go; router internal/jackettapi/router.go. Owns Jackett credentials + indexer overrides + autoconfig run history; encrypted SQLite at /config/boba.db.

3a. HTTP endpoints (internal/jackettapi/router.go + handlers)

Feature	Impl	Wired	Tests	Validation	Video
GET /healthz (router.go:60, health.go)	implemented	yes	go jackettapi/ health_test.go	unit (go)	N/A (no UI â€” go-test-covered endpoint)
GET /openapi.json (router.go:64, openapi.go)	implemented	yes	go jackettapi/ openapi_test.go; contract tests/contract/ openapi_test.go	unit + contract (go)	N/A (no UI â€” go-test + contract-covered endpoint)
GET /api/v1/jackett/credentials â€” list (router.go:67, credentials.go)	implemented	yes	go credentials_test.go; integration tests/integration/ jackett_db_test.go	unit + integration (go)	N/A (no UI â€” test-covered + exercised web /jackett/credentials page)
POST /api/v1/jackett/credentials â€” create/update (router.go:67, credentials.go)	implemented	yes	go credentials_test.go	unit + integration (go)	N/A (no UI â€” test-covered + exercised web /jackett/credentials page)
DELETE /api/v1/jackett/credentials/{name} (router.go:77, credentials.go)	implemented	yes	go credentials_test.go	unit + integration (go)	N/A (no UI â€” test-covered + exercised web /jackett/credentials/{name} page)

Feature	Impl	Wired Tests	Validation	Video
GET /api/v1/jackett/indexers â€” list configured (router.go:86, indexers.go)	implemented yes	go jackettapi/indexers_test.go	unit (go)	credential page) N/A (no U â€” test-covered + exercised web /jack indexers page)
POST/PATCH/DELETE /api/v1/jackett/indexers/{id} (router.go:93, indexers.go)	implemented yes	go indexers_test.go	unit (go)	N/A (no U â€” test-covered + exercised web /jack indexers page)
POST /api/v1/jackett/indexers/{id}/test â€” test config (indexers.go:239)	implemented yes	go indexers_test.go	unit (go)	N/A (no U â€” test-covered + exercised web /jack indexers page)
GET /api/v1/jackett/catalog â€” indexer catalog (router.go:112, catalog.go)	implemented yes	go jackettapi/catalog_test.go; benchmark catalog_bench_test.go	unit + bench (go)	N/A (no U â€” test-covered + exercised web /jack catalog ta
POST /api/v1/jackett/catalog/refresh (router.go:119, catalog.go)	implemented yes	go catalog_test.go	unit (go)	N/A (no U â€” test-covered + exercised web /jack catalog ta
GET /api/v1/jackett/autoconfig/runs â€” run history (router.go:137, runs.go)	implemented yes	go jackettapi/runs_test.go; e2e tests/e2e/jackett_management_test.go	unit + e2e (go)	N/A (no U â€” test-covered + exercised web /jack History ta
GET /api/v1/jackett/autoconfig/runs/{id} â€” run detail (router.go:144, runs.go)	implemented yes	go runs_test.go	unit (go)	N/A (no U â€” test-covered + exercised web /jack History ta
POST /api/v1/jackett/autoconfig/run â€” trigger run	implemented yes	go runs_test.go; e2e tests/e2e/jackett_management_test.go	unit + e2e (go)	N/A (no U â€” test-covered + exercised

Feature	Impl	Wired Tests	Validation	Video
(router.go:151, runs.go) GET/POST /api/v1/jackett/overrides (router.go:160, overrides.go) DELETE /api/v1/jackett/overrides/{env} (router.go:170, overrides.go)	implemented	yes go jackettapi/overrides_test.go	unit (go)	web /jackett History tab N/A (no UI) â€” go-test-covered endpoint)
Admin auth middleware (admin/admin on mutating routes) (jackettapi/auth_middleware.go)	implemented	yes go overrides_test.go	unit (go)	N/A (no UI) â€” go-test-covered endpoint)
Hardened CORS middleware (jackettapi/cors_middleware.go)	implemented	yes go cors_middleware_test.go	unit (go)	N/A (no UI) â€” go-test-covered middleware)

3b. Autoconfig engine + persistence (internal/jackett/, internal/db/)

Feature	Impl	Wired Tests	Validation	Video
Jackett autoconfig engine (jackett/autoconfig.go)	implemented	yes go jackett/autoconfig_test.go, autoconfig_bench_test.go	unit + bench (go)	N/A (no UI) â€” go-test-covered engine; exercised by web /jackett autoconfig run)
Fuzzy indexer matcher (jackett/matcher.go)	implemented	yes go jackett/matcher_test.go	unit (go)	N/A (no UI) â€” go-test-covered matcher)
Jackett HTTP client (jackett/client.go)	implemented	yes go jackett/client_test.go	unit (go)	N/A (no UI) â€” go-test-covered client)
Encrypted SQLite â€” AES-256-GCM crypto (db/crypto.go)	implemented	yes go db/crypto_test.go; security tests/security/credential_leak_test.go	unit + security (go)	N/A (no UI) â€” go-test + security-covered crypto)
	implemented	yes go db/migrate_test.go	unit (go)	

Feature	Impl	Wired Tests	Validation	Video
SQLite migrations (db/ migrate.go)				N/A (no UI â€” go- test- covered migrations)
Credential / indexer / runs repos (db/repos/ *.go)	implemented	yes go db/repos/*_test.go	unit (go)	N/A (no UI â€” go- test- covered repos)
.env bootstrap import (bootstrap/ *.go)	implemented	yes go bootstrap/ bootstrap_test.go	unit (go)	N/A (no UI â€” go- test- covered bootstrap)
Master-key ensure (first-boot) (bootstrap/ *.go)	implemented	yes go bootstrap/ bootstrap_test.go	unit (go)	N/A (no UI â€” go- test- covered first-boot key ensure)
.env file parse/ write (envfile/ *.go)	implemented	yes go envfile/parse_test.go, write_test.go	unit (go)	N/A (no UI â€” go- test- covered env file I/ O)

4. Tracker plugins (plugins/*.py)

Plugin contract: class with url, name, supported_categories, search(), download_torrent(). Installed into container by install-plugin.sh. The install-plugin.sh PLUGINS array names **44 plugins** (curated install set); the plugins/ tree carries **27 *.py files** (24 tracker/aggregator plugins + 3 support: nova2, novaprinter, socks, helpers, env_loader, theme_injector, download_proxy). Many curated names have NO matching file in this repo tree (the install script fetches/expects them elsewhere) â€” flagged honestly below.

4a. Plugins present in plugins/ AND named in the curated array

Plugin	Type	Impl	Wired (curated?)	Tests
eztv.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_

Plugin	Type	Impl	Wired (curated?)	Tests
limetorrents.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_ word tests/unit/ test_plugin_multiword_query_enc chaos tests/stress_chaos/ test_multiword_query_encoding_ (Â§11.4.85)
piratebay.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
solidtorrents.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
torlock.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
rutor.py	public (anonymous)	implemented	yes (curated)	stress test_plugin_parsers_stress_

Plugin	Type	Impl	Wired (curated?)	Tests
nyaa.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
anilibra.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
bitsearch.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
gamestorrents.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
kickass.py	public	implemented (degrades to honest empty)	yes (curated)	stress test_plugin_parsers_stress_

Plugin	Type	Impl	Wired (curated?)	Tests
megapeer.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
tokyotoshokan.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
torrentgalaxy.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
torrentkitty.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_
yts.py	public	implemented	yes (curated)	stress test_plugin_parsers_stress_

Plugin	Type	Impl	Wired (curated?)	Tests
jackett.py	aggregator	implemented	yes (curated)	covered via jackett autoconfig tests
iptorrents.py	private (freeleech-only)	implemented	yes (curated)	integration tests/integration/test_t
rutracker.py	private (cookies / CAPTCHA)	implemented	yes (curated)	integration test_tracker_auth_live.p tests/unit/ rutracker-cookie test (Â§1.1); test_plugin_parsers_stress_chao Â§11.4.85)
kinozal.py	private	implemented	yes (curated)	integration test_tracker_auth_live.p
nnmclub.py	private (cookies)	implemented	yes (curated)	integration test_tracker_auth_live.p

Plugin	Type	Impl	Wired (curated?)	Tests
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4b. Plugin support modules

Module	Type	Impl	Wired	Tests	Validation	Video
<code>nova2.py</code> <code>nova3</code> search harness	support	implemented	yes	covered via search e2e	tested-green-in-suite	N/A (no UI <code>test-covered, search harness; exercised by web search flow</code>)
<code>novaprinter.py</code> <code>result emitter (NDJSON-patched)</code>	support	implemented	yes	covered via search e2e	tested-green-in-suite	N/A (no UI <code>test-covered, result emitter; exercised by web search flow</code>)
<code>helpers.py</code> <code>HTTP/retrieve helpers</code>	support	implemented	yes	covered via search e2e	tested-green-in-suite	N/A (no UI <code>test-covered helper; exercised by web search flow</code>)
<code>socks.py</code> <code>SOCKS proxy support</code>	support	implemented	yes	covered indirectly	not-validated	N/A (no UI <code>internal SOCKS support; covered indirectly</code>)
<code>env_loader.py</code> <code>env/credential loader for plugins</code>	support	implemented	yes	covered via tracker auth tests	tested-green-in-suite	N/A (no UI <code>test-covered credential loader; exercised</code>)

Module	Type	Impl	Wired Tests	Validation	Video	
Per-plugin download_torrent() path (.torrent fetch)	feature	implemented	yes	integration test_buttons_api.py (download/file)	tested- green-in- suite	by web Trackers tab) N/A (no UI “” test- covered + exercised by web result-row Download button) N/A (no UI “” test- covered + exercised by web result-row Download (freeleech) policy)
Freeleech-only download policy (IPTorrents [free] tag)	policy	implemented	yes	integration test_iptorrents.py	tested- green-in- suite	

4c. Discrepancy rows (honest, §11.4.6)

Plugin	Type	Impl	Wired (curated?)	Tests	Validation	Video
torrentproject / torrentscsv	public (curated names)	named in install- plugin.sh PLUGINS array but NO matching plugins/ *.py file in tree	curated- name-only	none	not- implemented- in-repo-tree (install script expects them elsewhere)	N/A (not implemented in repo tree “ curated name, no plugin file)
23 other curated names absent from tree: academictorrents, ali213, audiobookbay, bitru, bt4g, btsow, extratorrent, glotorrents, linuxtracker, one337x, pctorrent, pirateiro, rockbox, snowfl, therarbg,	public (curated names)	named in curated array; NO plugins/ *.py in this tree	curated- name-only	none	not- implemented- in-repo-tree (name in install array; file absent)	N/A (not implemented in repo tree “ curated names, no plugin files)

Plugin	Type	Impl	Wired (curated?)	Tests	Validation	Video
torrentdownload, torrentfunk, xfsusb, yihua, yourbittorrent (+ torrentproject, torrentscsv)						

NOTE (Â§11.4.6, CORRECTION vs Rev 2): the `install-plugin.sh` `PLUGINS` array contains **44 entries**, not 12. Of those, ~21 have a matching `plugins/*.py` file in this repo; the rest are curated names with no file present in the tree. Stated as fact, not asserted as working.

5. Angular 21 frontend dashboard (frontend/)

Standalone SPA, signals-based; built to `download-proxy/src/ui/dist/frontend`, served from `:7187`. Vitest unit tests; Playwright e2e under `frontend/e2e`.

5a. Dashboard component controls (components/dashboard/dashboard.component.ts)

Feature	Impl	Wired Tests	Val
Search box (query signal) + submit	implemented	yes vitest dashboard.component.spec.ts; py integration test_dashboard_rendering.py, test_ui_comprehensive.py	test gre suit
Result grid “sortable (name/type/ size/seeds/leechers/quality/sources)	implemented	yes vitest dashboard.component.spec.ts	unit (vit
Results tab	implemented	yes py test_ui_comprehensive.py	test gre suit

Feature	Impl	Wired Tests	Val
Active Downloads tab + view (<code>dashboard.component.ts:745 loadDownloads</code>)	implemented yes	py <code>test_buttons_api.py</code>	test gre suit
Trackers tab + per-tracker auth chips (expandable)	implemented yes	py <code>test_auth_state_ui.py</code>	test gre suit
Schedules tab	implemented yes	py <code>test_dashboard_automation.py</code>	test gre suit
Hooks tab	implemented yes	py <code>test_dashboard_automation.py</code>	test gre suit
Per-row Magnet button â†’ magnet dialog trigger	implemented yes	vitest <code>magnet- dialog.component.spec.ts</code> ; py <code>test_magnet_dialog.py</code>	test gre suit
Per-row add-to-qBittorrent (qBit) button	implemented yes	py <code>test_button_functions.py</code> , <code>test_buttons_api.py</code>	test gre suit
Per-row Download (.torrent) button	implemented yes	py <code>test_buttons_api.py</code>	test gre suit
	implemented yes		

Feature	Impl	Wired Tests	Validation
Per-row processing spinner (isBusy())		vitest dashboard.component.spec.ts	unit (vitest)
Bridge health indicator + retry button	implemented yes	py test_bridge_root_liveness.py	test green-in-suite
Tracker-stat dialog trigger	implemented yes	py contract test_tracker_stats_contract.py	test green-in-suite

5b. Dialog + UI components

Feature	Impl	Wired Tests	Validation
Magnet dialog “readonly textarea + Copy/Open/Add/Close (magnet- dialog.component.ts)	implemented yes	vitest magnet- dialog.component.spec.ts; py test_magnet_dialog.py	tested- green-in- suite
qBit login dialog “user/pass/ remember/Login/Cancel (qbit- login- dialog.component.ts)	implemented yes	vitest qbit-login- dialog.component.spec.ts; py test_login_actions.py, test_auth_state_ui.py	tested- green-in- suite
Theme picker “palette dropdown + light/dark toggle (theme- picker.component.ts)	implemented yes	vitest theme- picker.component.spec.ts; py e2e test_theme_runtime.py, test_crossapp_theme.py	tested- green-in- suite
Confirm dialog “title/message/ Confirm/Cancel/backdrop (confirm- dialog.component.ts)	implemented yes implemented yes	vitest confirm- dialog.component.spec.ts	unit (vitest)

Feature	Impl	Wired Tests	Validation
Tracker-stat dialog â€” JSON view + Copy + status badge + notes table + Esc-close (tracker-stat-dialog.component.ts)		vitest tracker-stat-dialog.component.spec.ts; py contract test_tracker_stats_contract.py	tested-green-in-suite
Toast container â€” info/success/warning/error auto-dismiss (toast-container.component.ts)	implemented yes	vitest toast-container.component.spec.ts	unit (vitest)
Site footer (site-footer.component.ts)	implemented yes	vitest	unit (vitest)

5c. Services (frontend/src/app/services/)

Feature	Impl	Wired Tests	Validation	Video
API service â€” HTTP wrapper (search/download/magnet/auth/schedules/hooks/downloads/stats/config), 15s timeout, runtime baseUrl (api.service.ts)	implemented yes	vitest api.service.spec.ts	unit (vitest)	N/A (no U â€” TS service; exercised every web dashboard call)
SSE service â€” EventSource live result streaming + bearer-token (sse.service.ts)	implemented yes	vitest sse.service.spec.ts; py integration test_realtime_streaming.py, test_streaming_browser.py	tested-green-in-suite	N/A (no U â€” TS service; exercised web search flow live- results stream)
Theme service â€” palette/ mode + localStorage + cross-app sync (theme.service.ts)	implemented yes	vitest theme.service.spec.ts; py e2e test_theme_runtime.py, test_crossapp_theme.py	tested-green-in-suite	N/A (no U â€” TS service; exercised web them picker)
Toast service â€” toast queue state (toast.service.ts)	implemented yes	vitest toast.service.spec.ts	unit (vitest)	N/A (no U â€” TS service; exercised web toast notificatio
Dialog service â€” promise-based confirm state (dialog.service.ts)	implemented yes	vitest dialog.service.spec.ts	unit (vitest)	N/A (no U â€” TS service; exercised web confi dialogs)

5d. Jackett page (frontend/src/app/jackett/)

Feature	Impl	Wired Tests	Validation
Credentials â€” list + Add/Edit/Delete (per-row, window.confirm) + banners (jackett/credentials/credentials.component.ts)	implemented yes	vitest credentials.component.spec.ts	unit (vitest)
Credential edit dialog â€” name/username/password/cookies/Save/Cancel (credential-edit-dialog.component.ts)	implemented yes	vitest credential-edit-dialog.component.spec.ts	unit (vitest)
Indexers â€” Configured/Catalog/History tab container (jackett/indexers/indexers.component.ts)	implemented yes	vitest indexers.component.spec.ts	unit (vitest)
Indexers â€” Configured tab (list + per-row remove)	implemented yes	vitest indexers.component.spec.ts	unit (vitest)
Indexers â€” Catalog tab (list + per-row add)	implemented yes	vitest indexers.component.spec.ts	unit (vitest)
Indexers â€” History tab (autoconfig run history)	implemented yes	vitest indexers.component.spec.ts	unit (vitest)
Indexer add dialog â€” credential dropdown + Save/Cancel + error banner (indexer-add-dialog.component.ts)	implemented yes	vitest indexer-add-dialog.component.spec.ts	unit (vitest)
IPorrents cookie-flow component (iptorrents-cookie-flow.component.ts)	implemented yes	vitest (jackett specs)	unit (vitest)

5e. Frontend infra

Feature	Impl	Wired	Tests	Validation	Video
Frontend coverage thresholds (40% lines/branches/funcs/stmts, v8)	infra	yes	vitest.config.ts	enforced in vitest config	N/A (no UI â€” build/coverage infra)

6. BobaLink browser extension (extension/)

WXT + TypeScript Manifest-V3. Detects magnet/.torrent links and forwards to merge service :7187. Status detail: docs/browser_extension/Status.md. Built zips at extension/.output/bobalink-1.0.0-{chrome,firefox}.zip.

6a. User-facing surfaces

Feature	Impl	Wired Tests	Validation	Video
Popup “ per-row Send button (src/popup/, entypoints/popup.ts)	implemented yes	ally tests/ally/popup.ally.test.ts	tested-green-in-suite	VIDEO-CONFIRMED 2026-06-16 “ boba-extension-popup.mp4; verdict docs/qa/recordings-202606 boba-extension-verdict.md
Popup “ Send-All button	implemented yes	ally popup.ally.test.ts	tested-green-in-suite	VIDEO-CONFIRMED 2026-06-16 “ boba-extension-popup.mp4; verdict docs/qa/recordings-202606 boba-extension-verdict.md
Popup “ Refresh button + status indicator (online/offline/warning)	implemented yes	ally popup.ally.test.ts	tested-green-in-suite	VIDEO-CONFIRMED 2026-06-16 “ boba-extension-popup.mp4; verdict docs/qa/recordings-202606 boba-extension-verdict.md
Popup “ torrent list (name + type + infohash prefix + sent badge)	implemented yes	ally popup.ally.test.ts	tested-green-in-suite	VIDEO-CONFIRMED 2026-06-16 “ boba-extension-popup.mp4; verdict docs/qa/recordings-202606 boba-extension-verdict.md
Popup “ Open Options link	implemented yes	ally popup.ally.test.ts	tested-green-in-suite	VIDEO-CONFIRMED 2026-06-16 “ boba-extension-popup.mp4; verdict docs/qa/recordings-202606 boba-extension-verdict.md
Options page (src/options/options.ts)	implemented yes	ally options.ally.test.ts; integration options-background.test.ts	tested-green-in-suite	VIDEO-CONFIRMED 2026-06-16 “ boba-extension-options.mp4; verdict docs/qa/recordings-202606

Feature	Impl	Wired Tests	Validation	Video
Content script â€” link/text highlight overlay (src/content/highlight.ts, entrypoints/content.ts)	implemented yes	integration content-background.test.ts; security content-xss.test.ts	tested-green-in-suite	boba-extension-verdict.md VIDEO-CONFIRMED 2026-06-16 â€” boba-extension-scan-detect.mp4; verdict docs/qa/recordings-202606 boba-extension-verdict.md
Background service worker â€” message hub (src/background/, entrypoints/background.ts)	implemented yes	integration tests/integration/*-background.test.ts	tested-green-in-suite	N/A (no UI â€” test-cover + exercised by extension scan/popup)
Context-menu entries (right-click torrent actions)	implemented yes	integration *-background.test.ts	tested-green-in-suite	N/A (no UI â€” test-cover + exercised by extension scan/popup)
Keyboard shortcuts (scan-now / highlight-toggle / send-all)	implemented yes	integration *-background.test.ts	tested-green-in-suite	N/A (no UI â€” test-cover + exercised by extension scan/popup)
Action badge (detected count) + notifications	implemented yes	integration *-background.test.ts	tested-green-in-suite	N/A (no UI â€” test-cover + exercised by extension scan/popup)

6b. Scanners (src/scanner/)

Feature	Impl	Wired Tests	Validation	Video
Base scanner â€” computeStableId() lifecycle (scanner/base.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised by extension scan-detect)
Link scanner â€” <a href> magnet/.torrent (scanner/link-scanner.ts)	implemented yes	security scanner-hostile-input.test.ts; perf scanner.perf.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised by extension scan-detect)

Feature	Impl	Wired Tests	Validation	Video
Text scanner â€” bare magnet text (scanner/text-scanner.ts)	implemented yes	security scanner-hostile-input.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised by extension scan-detect)
Scanner orchestrator â€” dedup + mutation-observer re-scan (scanner/orchestrator.ts)	implemented yes	perorchestrator-scaling.perf.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised by extension scan-detect)
Site DB â€” per-site selector match rules (scanner/site-db.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised by extension scan-detect)

6c. Parsers (src/parser/)

Feature	Impl	Wired Tests	Validation	Video
Bencode parser (parser/bencode.ts)	implemented yes	security bencode-torrentfile-hostile.test.ts; perf parsers.perf.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered parser; exercised by extension scan-detect)
Magnet parser (parser/magnet.ts)	implemented yes	security infohash-detection-hostile.test.ts; perf magnet.perf.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered parser; exercised by extension

Feature	Impl	Wired	Tests	Validation	Video
Torrent-file parser (parser/torrent-file.ts)	implemented	yes	security bencode-torrentfile-hostile.test.ts	tested-green-in-suite	scan-detect) N/A (no UI â€” test-covered parser; exercised by extension scan-detect)

6d. API client + offline (src/api/)

Feature	Impl	Wired	Tests	Validation	Video
Boba API client â€” POST /api/v1/download, retry + rate-limit (api/boba-client.ts)	implemented	yes	vitest unit; chaos tests/chaos/boba-client-resilience.chaos.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered API client; exercised by extension Send)
Health probe â€” probeHealth() (api/health.ts)	implemented	yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered health probe; exercised by extension status indicator)
Offline send queue â€” IndexedDB enqueue + re-dispatch (api/queue.ts)	implemented	yes	chaos tests/chaos/queue.chaos.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered offline queue; chaos-covered)

6e. Shared utilities + tabgroups (src/shared/, src/tabgroups/)

Feature	Impl	Wired	Tests	Validation	Video
	implemented	yes	security crypto-tamper.test.ts,		N/A (no UI â€”

Feature	Impl	Wired Tests	Validation	Video
Crypto â€” AES-256-GCM token encrypt/decrypt (shared/crypto.ts)		no-hardcoded-secret.test.ts	tested-green-in-suite	test-covered crypto; exercised by extension Send token path)
Storage â€” namespaced chrome.storage.local wrapper (shared/storage.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered storage wrapper)
Constants â€” site selectors / rate-limit / retry config (shared/constants.ts)	implemented yes	covered via scanner tests	tested-green-in-suite	N/A (no UI â€” test-covered constants)
Errors â€” Network/Server/Storage error types (shared/errors.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered error types)
Events â€” TypedEventEmitter bus (shared/events.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered event bus)
Logger â€” namespaced, no-secret (shared/logger.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered logger)
Utils â€” debounce / TokenBucket rate limiter (shared/utils.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered utils/rate-limiter)
Tab groups â€” batch grouped-tab torrents (tabgroups/index.ts)	implemented yes	vitest unit (tab-group Challenge per ext Status)	tested-green-in-suite	N/A (no UI of its own â€” test-covered tab-group batching; exercised by

Feature	Impl	Wired Tests	Validation	Video extension Send-All)
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6f. i18n locales (extension/src/public/_locales/)

Feature	Impl	Wired	Tests	Validation	Video
Locale en (English)	implemented yes		i18n tests/ i18n/locale- safety.test.ts	tested-green-in- suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale de (German)	implemented yes		i18n locale- safety.test.ts	tested-green-in- suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale es (Spanish)	implemented yes		i18n locale- safety.test.ts	tested-green-in- suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale fr (French)	implemented yes		i18n locale- safety.test.ts	tested-green-in- suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
	implemented yes				

Feature	Impl	Wired	Tests	Validation	Video
Locale i t (Italian)			i18n locale- safety.test.ts suite	tested-green-in- suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale j a (Japanese)	implemented yes		i18n locale- safety.test.ts suite	tested-green-in- suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale p t (Portuguese)	implemented yes		i18n locale- safety.test.ts suite	tested-green-in- suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale ru (Russian)	implemented yes		i18n locale- safety.test.ts suite	tested-green-in- suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)

6g. Live round-trip

Feature	Impl	Wired Tests	Validation	Video
Live download endpoint round-trip (tests/live/download-endpoint.live.test.ts)	implemented	yes (live-gated) live download- endpoint.live.test.ts	partial â€” operator- gated live env	N/A (no UI â€” live round-

Feature	Impl	Wired Tests	Validation Video
			trip test; operator- gated live env)

7. WebUI bridge (webui-bridge.py) â€” host process :7188

Bridges qBittorrent WebUI with private-tracker auth. NOT a container.

Feature	Impl	Wired Tests	Validation Video
WebUI proxy + header rewrite (webui-bridge.py)	implemented host	integration test_webui_bridge.py, test_bridge_header_rewrite.py; stress test_bridge_stress_chaos.py	tested- green-in- suite
Private-tracker auth bridging (rutracker/kinozal/nnmclub/iptorrents)	implemented host	integration test_webui_bridge.py	partial â€” live-gated
Theme injection into bridged WebUI (plugins/theme_injector.py)	implemented host	integration test_bridge_theme_injection.py	tested- green-in- suite
Bridge root liveness (/)	implemented host	integration test_bridge_root_liveness.py	tested- green-in- suite

8. Infrastructure / CLI / shell scripts

8a. boba-ctl CLI orchestrator (Go: cmd/boba-ctl/main.go + wrapper scripts/boba-ctl.sh)

Feature	Impl	Wired Tests	Validation	Video
boba-ctl up â€” start stack (cmd/boba-ctl/main.go:32)	implemented host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite	N/A (CLI â€” go-test-covered; exercised by boba-ctl CLI demo, boba-ctl orchestrator-demo.mp4)
boba-ctl down â€” stop stack (main.go:34)	implemented host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite	N/A (CLI â€” go-test-covered; exercised by boba-ctl CLI demo, boba-ctl orchestrator-demo.mp4)
boba-ctl status â€” service status (main.go:36)	implemented host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite + LIVE	VIDEO-CONFIRMED 2026-06-16 â€” boba-ctl orchestrator-demo.mp4; verdict docs qa/recordings-20260616-boba-cli-verdict.m
boba-ctl health â€” health probe (main.go:38, cmd/boba-ctl/health_test.go)	implemented host	go cmd/boba-ctl/health_test.go	tested-green-in-suite + LIVE	VIDEO-CONFIRMED 2026-06-16 â€” boba-ctl orchestrator-demo.mp4
boba-ctl list â€” list services (main.go:40)	implemented host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite + LIVE	VIDEO-CONFIRMED 2026-06-16 â€” boba-ctl orchestrator-demo.mp4
boba-ctl.sh wrapper â€” compose-compatible shim (up/down/ps/config/pull + passthrough) (scripts/boba-ctl.sh)	implemented host	covered via boba-ctl go tests	tested-green-in-suite	N/A (CLI shim â€” test-covered; exercised by boba-ctl CLI demo)

8b. Lifecycle scripts (repo root)

Feature	Impl	Wired Tests	Validation	Video
start.sh â€” container bring-up + Jackett key extraction/injection	implemented host	covered via e2e test_fixtures_bring_up_services.py	tested-green-in-suite	N/A (CLI shim â€” test-covered; exercised by boba-ctl CLI demo)

Feature	Impl	Wired Tests	Validation	Video
<code>stop.sh</code> "stop/ remove/purge"	implemented host	none dedicated	not- validated	N/A scrip no U cove oper
<code>setup.sh</code> "one-time setup"	implemented host	none dedicated	not- validated	N/A scrip no U cove oper
<code>start-proxy.sh</code> "proxy container entry helper"	implemented host	none dedicated	not- validated	N/A scrip no U cove oper
<code>install- plugin.sh</code> "copy curated plugins to container engines"	implemented host	covered indirectly (e2e search)	tested- green-in- suite	N/A scrip no U cove oper
<code>init-qbit- password.sh</code> "qBit admin password bootstrap"	implemented host	none dedicated	not- validated	N/A scrip no U cove oper
<code>fix-qbit- password.sh</code> "qBit admin password repair"	implemented host	none dedicated	not- validated	N/A scrip no U cove oper
<code>setup-webui- bridge- service.sh</code> + <code>webui- bridge.service</code> "systemd unit for bridge"	implemented host	none dedicated	not- validated	N/A scrip no U cove oper

8c. Test / CI wrappers (repo root)

Feature	Impl	Wired	Tests	Validation	Video
<code>ci.sh</code> "manual CI (syntax+unit+integration+e2e+health)"	implemented host		self (is the gate)	tested-green- in-suite	N/A (host CI/test wrapper "no UI; is the gate)
<code>run-all-tests.sh</code> "full suite (hardcoded podman)"	implemented host		self	partial " fails on docker-only hosts (known gotcha)	N/A (host CI/test wrapper "no

Feature	Impl	Wired	Tests	Validation	Video
<code>test.sh</code> “quick validation wrapper”	implemented	host	self	tested-green-in-suite	UI; is the gate) N/A (host CI/test wrapper “no UI; is the gate) N/A (host CI/test wrapper “no UI; is the gate)
<code>test-all.sh</code> “full validation wrapper”	implemented	host	self	tested-green-in-suite	UI; is the gate) N/A (host CI/test wrapper “no UI; is the gate) N/A (host CI/test wrapper “no UI; is the gate)
<code>test-full.sh</code> “extended validation wrapper”	implemented	host	self	tested-green-in-suite	UI; is the gate)

8d. scripts/ helpers

Feature	Impl	Wired	Tests	Validation	Video
<code>scripts/add-submodules.sh</code> “submodule bootstrap”	implemented	host	none	dedicated	not-validated N/A (host script “no UI; test-covered/operational)
<code>scripts/audit-plugins.sh</code> “plugin discrepancy audit”	implemented	host	none	dedicated	not-validated N/A (host script “no UI; test-covered/operational)
<code>scripts/build-releases.sh</code> “build extension/Go release artifacts”	implemented	host	none	dedicated	not-validated N/A (host script “no UI; test-covered/operational)
<code>scripts/codegraph_validate.sh</code> “CodeGraph index validation (Â§11.4.78)”	implemented	host	self	tested-green-in-suite	N/A (host script “no UI; test-covered/operational)
<code>scripts/docs_chain.sh</code> “docs_chain engine wrapper (Â§11.4.106)”	implemented	host	self	tested-green-in-suite	N/A (host script “no UI; test-covered/operational)
	implemented	host		not-validated;	N/A (host script “no UI; test-covered/operational)

Feature	Impl	Wired Tests	Validation	Video
<code>scripts/ensure-macos-tunnel.sh</code> "0.0.0.0 SSH tunnel"		none dedicated (log <code>qa-results/tunnel-keepalive.log</code>)	LAN threat-model RW-05	no UI; test-covered/ operational)
<code>scripts/tunnel-keepalive.sh</code> "tunnel self-heal"	implemented host	none dedicated	not-validated	N/A (host script "no UI; test-covered/ operational)
<code>scripts/freeze-openapi.sh</code> "freeze boba-jackett OpenAPI snapshot"	implemented host	<code>contract tests/contract/openapi_test.go</code>	tested-green-in-suite	N/A (host script "no UI; test-covered/ operational)
<code>scripts/generate_markdown_exports.sh</code> "md" "html/pdf export" (11.4.65)	implemented host	self	tested-green-in-suite	N/A (host script "no UI; test-covered/ operational)
<code>scripts/helixqa.sh</code> "HelixQA harness driver"	implemented host	self	not-validated	N/A (host script "no UI; test-covered/ operational)
<code>scripts/opencode-helixqa.sh</code> "OpenCode HelixQA driver"	implemented host	self	not-validated	N/A (host script "no UI; test-covered/ operational)
<code>scripts/install_git_hooks.sh</code> "git hook installer" (11.4.75)	implemented host	self	tested-green-in-suite	N/A (host script "no UI; test-covered/ operational)
<code>scripts/pre_build_verification.sh</code> "pre-build gate sweep"	implemented host	self (is the gate)	tested-green-in-suite	N/A (host script "no UI; test-covered/ operational)
<code>scripts/pre_code_review.sh</code> "code-review gate" (11.4.125)	implemented host	self	tested-green-in-suite	N/A (host script "no UI; test-covered/ operational)
<code>scripts/run_all_challenges.sh</code> "challenge bank runner"	implemented host	self	not-validated	N/A (host script "no UI; test-covered/ operational)
<code>scripts/run-tests.sh</code> "test runner helper"	implemented host	self		N/A (host script "

Feature	Impl	Wired Tests	Validation	Video
			tested-green-in-suite	no UI; test-covered/operational)
scripts/scan.sh "repo scan helper	implemented	host none dedicated	not-validated	N/A (host script "no UI; test-covered/operational)
scripts/track-build-resources.sh "build resource sampler (§11.4.24)	implemented	host none dedicated	not-validated	N/A (host script "no UI; test-covered/operational)

8e. Infra / compose / startup

Feature	Impl	Wired Tests	Validation	Video
docker-compose.yml "2-container default + Go profile + boba-jackett service	infra	yes e2e test_live_containers.py	tested-green-in-suite	N/A (in "no e2e-test-covered/orchest
Jackett auto-configuration at startup (key extract "inject JACKETT_API_KEY)	implemented	yes integration test_jackett_autoconfig_real.py	tested-green-in-suite	N/A (in "no integra test-co startup autocor
BOBA_MASTER_KEY auto-generation (first-boot) for encrypted boba.db	implemented	yes go bootstrap/bootstrap_test.go	unit (go)	N/A (in "no go-test covered boot ke generat

Components NOT inventoried / discrepancies (honest gaps, §11.4.6)

- **CORRECTION vs Rev 2:** boba-ctl.sh DOES exist at scripts/boba-ctl.sh (Rev 2 said it did not). Its subcommands are now itemized in §8a. The underlying Go binary is cmd/boba-ctl/main.go (5 subcommands: up/down/status/health/list).
- **CORRECTION vs Rev 2:** the install-plugin.sh PLUGINS curated array has **44 entries**, not 12. ~21 have a matching plugins/*.py file; the rest (incl.Â torrentproject, torrentscsv) are curated names with no file present in this tree " itemized in §4c, not asserted as working.
- **Prometheus/OpenTelemetry metrics** " tests/observability/test_metrics_exist.py + observability/ dir exist, but no prometheus_client Counter()/Histogram()/Gauge() definitions were found in download-proxy/src; metrics surface is via GET /api/v1/stats

(counters maintained in-process), not a `/metrics` Prometheus endpoint. Stated as fact, not assumed.

- **Go DownloadHandler / ActiveDownloadsHandler** – flagged partial/stub in §2a (mock success / empty array, no qBit integration) per source reading. The Go profile is opt-in and NOT the shipped product.
- **Go scheduler** – CRUD-only, no driver loop (RW-10); schedules never fire. The Go enricher is missing (RW-11).
- **Video-recording confirmation** – PENDING for all rows EXCEPT the boba-ctl CLI subcommands (status/health/list) and the web-UI flows already filmed (POST `/api/v1/search` journey + dashboard tab/button/theme/Jackett tour). Remaining per-feature flows are an in-progress recording pass (§11.4.107/§11.4.143).